Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-8296
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram

The diagram below maps the Credit Card Approval Prediction system onto the Presentation, Application, and Data layers.

Presentation / Client Layer

Flask-rendered Web App (HTML5, CSS3, Bootstrap, JavaScript) used by credit analysts, compliance officers, and prospective customers

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API Gateway / Load Balancer

Flask (Werkzeug WSGI server) routes requests between the UI and prediction service; exposes IBM Watson Machine Learning REST scoring endpoint for cloud inference

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Auth Service	Core Logic Service	External API Integrations
IBM Cloud IAM authentication for the Watson ML deployment; session-based access control on the Flask app	*Data preprocessing, feature engineering (binary risk-label conversion), and prediction using Logistic Regression, Random Forest, XGBoost, and Decision Tree models*	*IBM Watson Machine Learning API for scalable, real-time cloud-hosted approval predictions*

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Data / Storage Layer

Historical labeled applicant dataset (CSV/Pandas) used for training and evaluation; serialized best-performing model (Pickle/Joblib) persisted for inference; optional IBM Cloud Object Storage for dataset and model artifacts

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	A Flask-rendered web interface where credit analysts, compliance officers, and prospective customers enter applicant details (income, employment, credit history, loan balances) and instantly view the model's approve/reject prediction.	Flask (Jinja2 templates), HTML5, CSS3, Bootstrap, JavaScript
API Gateway	Routes incoming form/API requests to the prediction service and, for cloud deployments, forwards scoring requests to the IBM Watson Machine Learning endpoint, returning the prediction to the client.	Flask / Werkzeug WSGI server, IBM Watson Machine Learning REST API
Microservices (Core Logic Service)	Cleans and encodes applicant inputs, converts multi-class payment status codes into binary high-risk/low-risk labels, and runs the trained classification model to generate the approval decision.	Python, Pandas, NumPy, Scikit-learn, XGBoost, Decision Tree, Random Forest, Logistic Regression
Database	Stores the historical labeled credit card applicant records used to train and evaluate the four models, and persists the serialized best-performing model artifact used at prediction time.	CSV dataset (Pandas DataFrame), Pickle/Joblib model file, IBM Cloud Object Storage (optional)